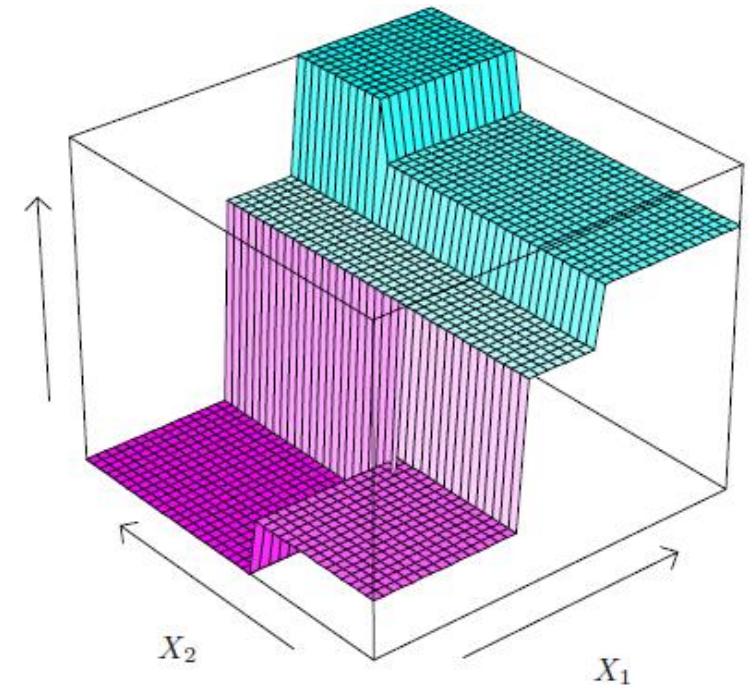
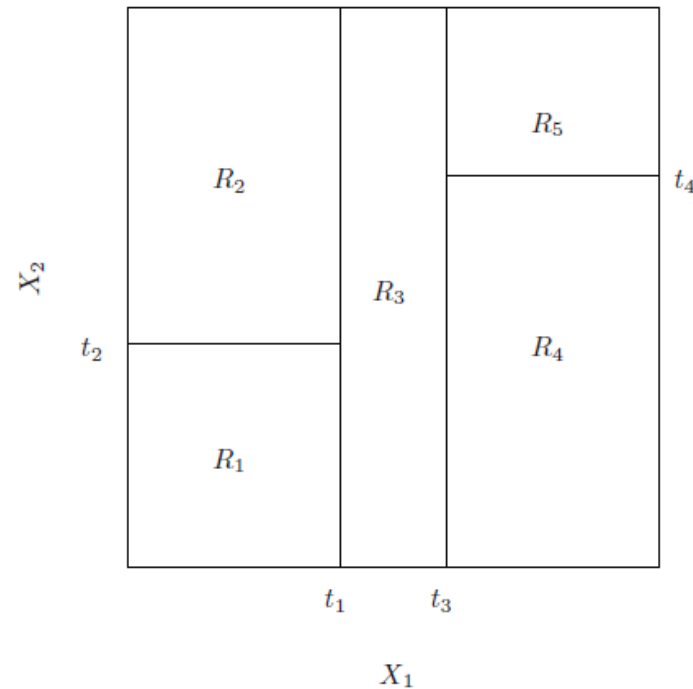
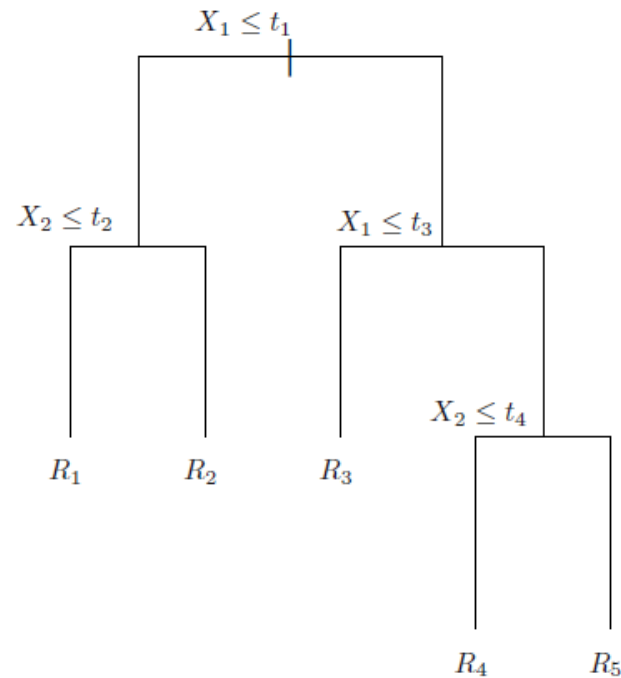


Week 14 Decision Tree

2026.05.28

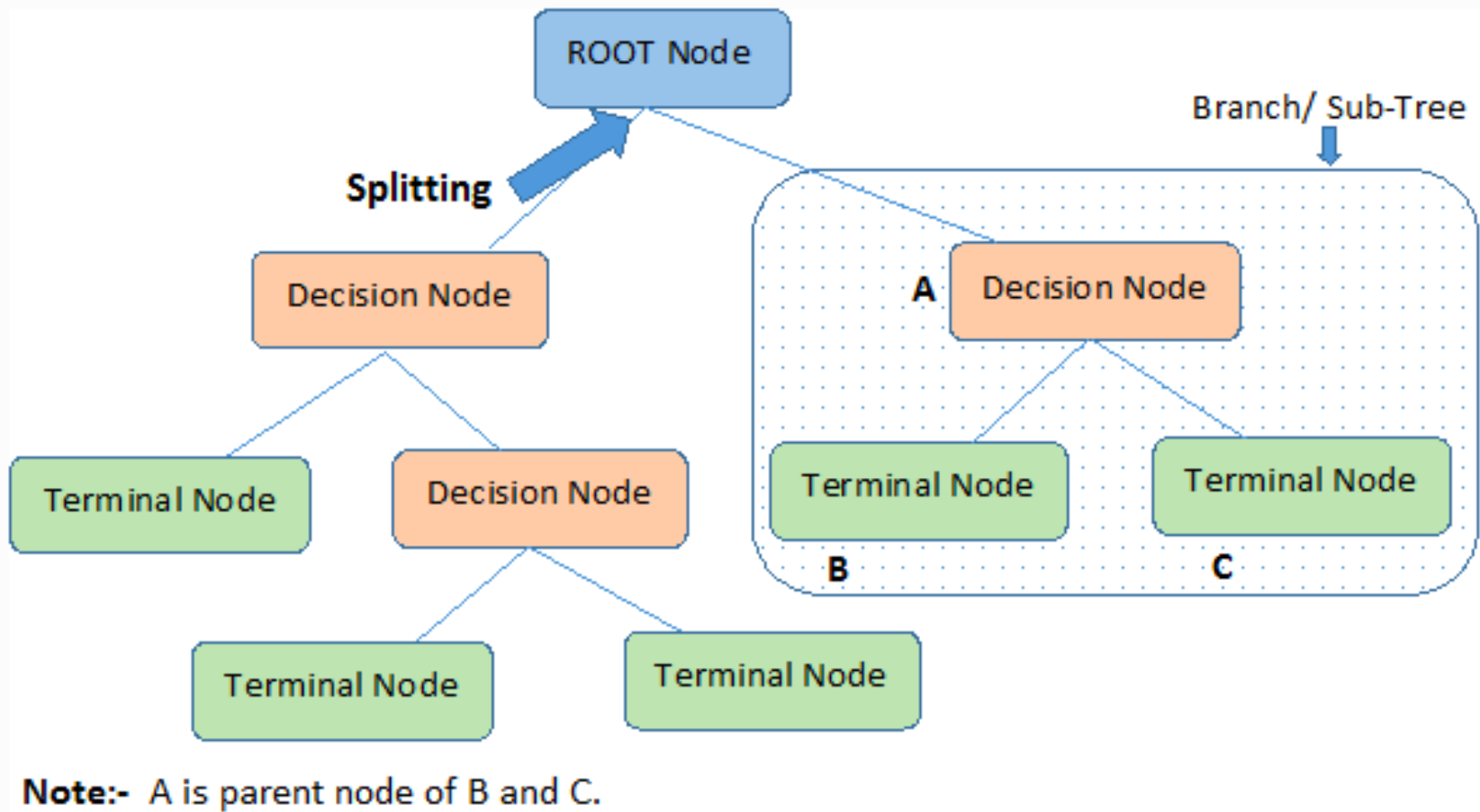
Tree-Based Methods

- Partition the feature space into a set of rectangles.



Tree-Based Methods

- Conceptually simple, powerful
- Interpretability: it mimics the way people think.
- Decision Trees (決策樹), an approach used in data mining/machine learning
- CART: classification and regression tree
 - Y continuous: regression tree
 - Y discrete: classification tree



Node (節點):

- Root (node)
- Decision node/internal node
- Leaf (node)/terminal node

Branch (分支)

Grow (種植) a tree

Splitting (切開、劃分)

Recursive binary partition

Grow a tree

1. Decide the splitting variable and the split point

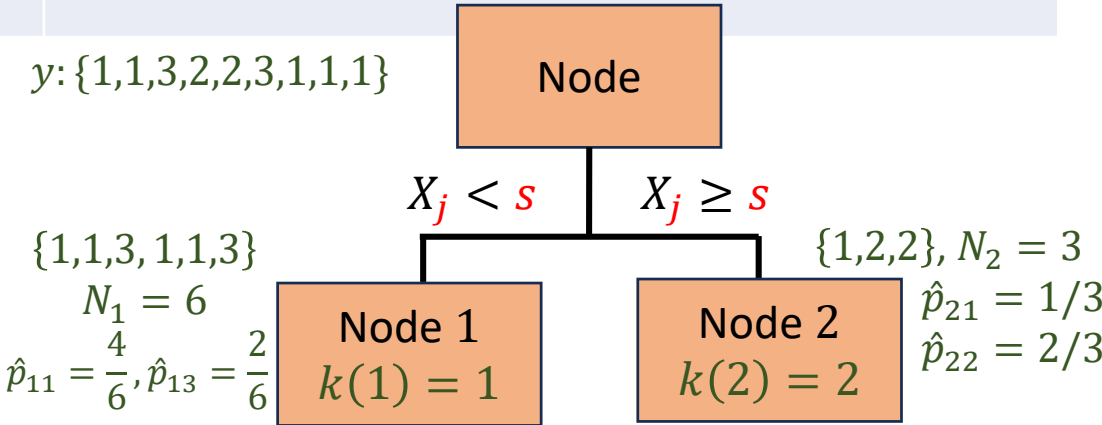
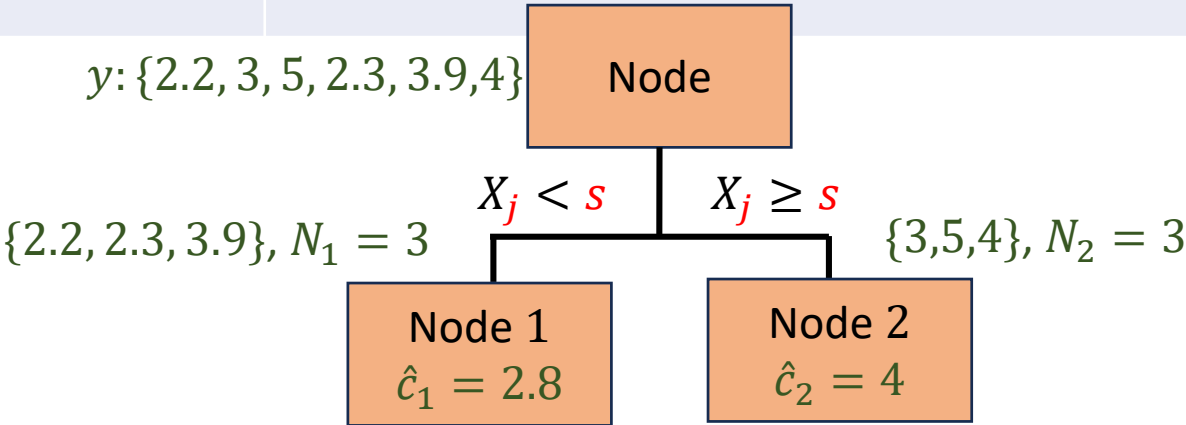
- Node impurity measure: objective function, minimize the impurity

2. Decide the tree size

- A very large tree might overfit the data
- A small tree might not capture the important structure
- Ex 1: split tree nodes only if the decrease in sum-of-squares due to the split exceeds some threshold.
- Ex 2: grow a large tree (T_0) (stop the splitting process only when some minimum node size (say 5) is reached) $\Rightarrow$ tree then is pruned using cost-complexity pruning

Grow a tree: 1. splitting variable j and split point s

CART	Regression Tree	Classification tree
Response Y	Continuous	Categorical: {1,2,...,K}
Prediction for node m	$\hat{c}_m = \bar{y}_m = \frac{1}{N_m} \sum_{x_i \in R_m} y_i$ (average value)	$k(m) = \arg \max \hat{p}_{mk},$ where $\hat{p}_{mk} = \frac{1}{N_m} \sum_{x_i \in R_m} I(y_i = k)$ (max proportion)
Impurity measure of node m, $Q_m(T)$	$\frac{1}{N_m} \sum_{x_i \in R_m} (y_i - \hat{c}_m)^2$ (prediction error)	$1 - \hat{p}_m k(m)$ (misclassification error)
Splitting	$(j, s) = \arg \min \left[\sum_{x_i \in R_1} (y_i - \hat{c}_1)^2 + \sum_{x_i \in R_2} (y_i - \hat{c}_2)^2 \right]$	$(j, s) = \arg \min [N_1(1 - \hat{p}_{1k(1)}) + N_2(1 - \hat{p}_{2k(2)})]$



Grow a tree: 1. splitting variable j and split point s

CART

Regression Tree

Classification tree

Categorical: {1,2,...,K}

Measurement of **node purity**- two other measures :

- Gini index

$$G = \sum_{k=1}^K \hat{p}_{mk}(1 - \hat{p}_{mk})$$

- Entropy

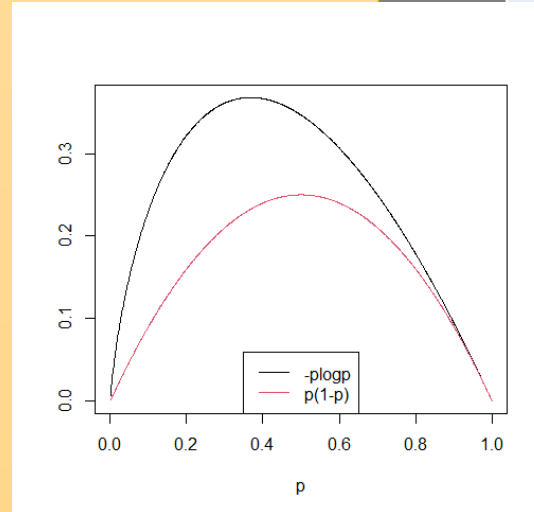
$$D = - \sum_{k=1}^K \hat{p}_{mk} \log \hat{p}_{mk}$$

Ex: Node1

$$G = \frac{4}{6} \left(1 - \frac{4}{6}\right) + 0(1 - 0) + \frac{2}{6} \left(1 - \frac{2}{6}\right) = 0.44$$

$$D = - \left(\frac{4}{6} \log \frac{4}{6} + 0 + \frac{2}{6} \log \frac{2}{6} \right) = 0.64$$

$$E = 1 - \frac{4}{6} = 0.33 \text{ (Classification error rate)}$$



$$k(m) = \arg \max \hat{p}_{mk},$$

$$\text{where } \hat{p}_{mk} = \frac{1}{N_m} \sum_{x_i \in R_m} I(y_i = k)$$

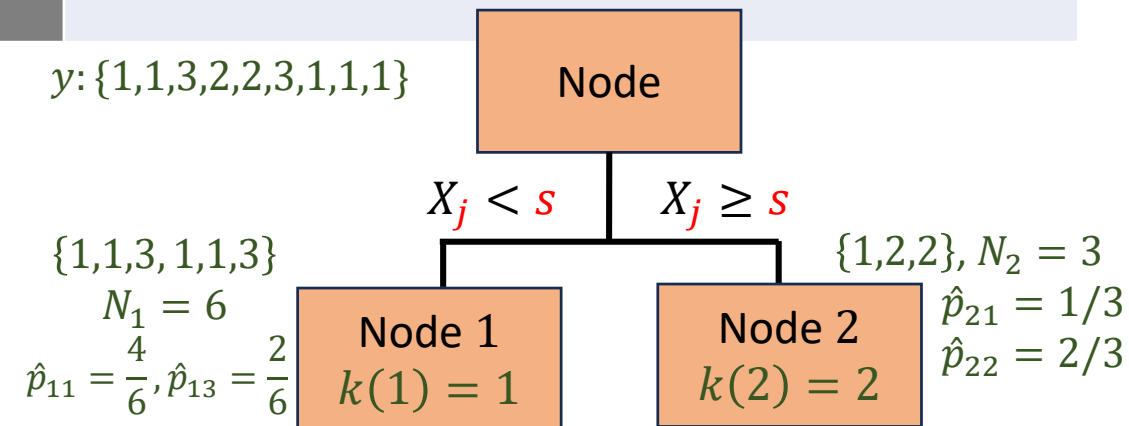
(max proportion)

$$1 - \hat{p}_m k(m)$$

(misclassification error)

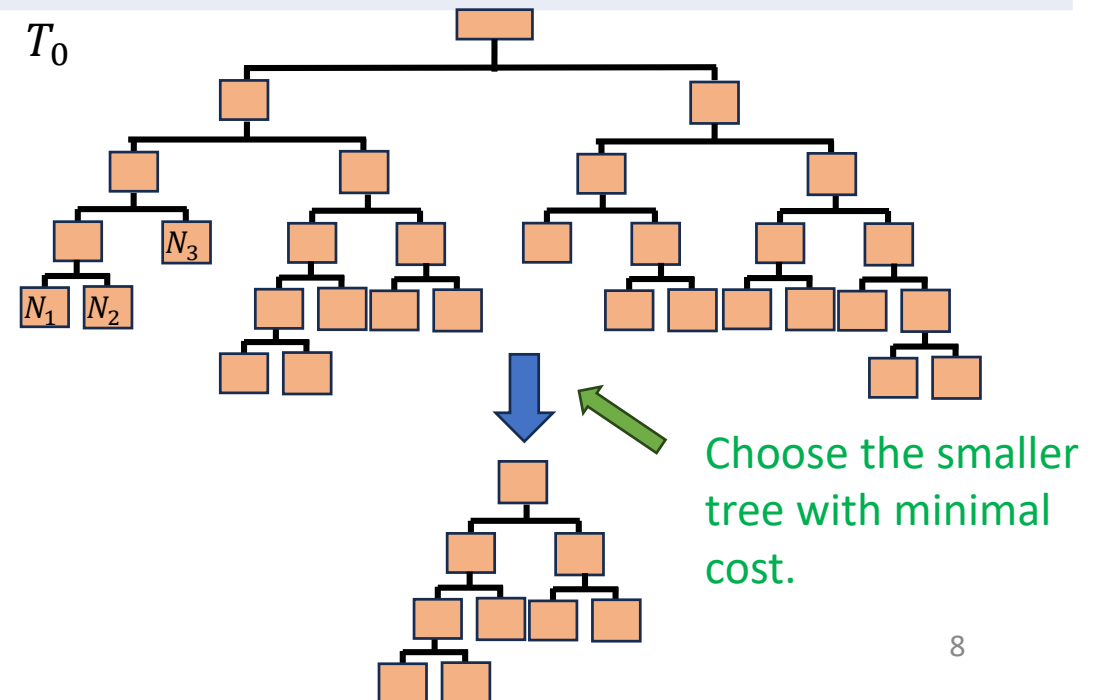
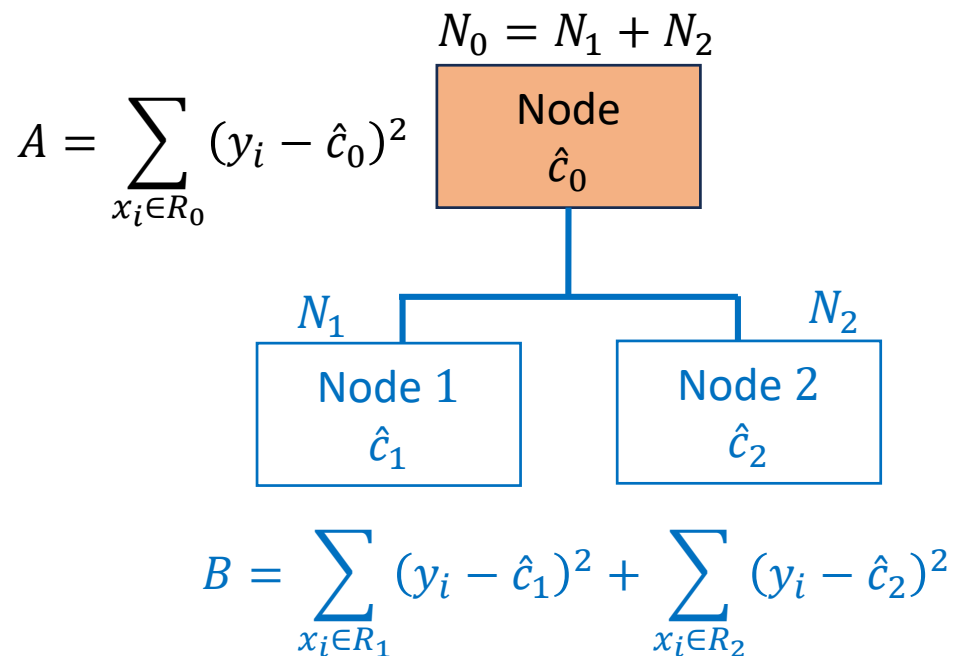
$$(j, s) = \arg \min [N_1(1 - \hat{p}_{1k(1)}) + N_2(1 - \hat{p}_{2k(2)})]$$

y: {1,1,3,2,2,3,1,1,1}



Grow a tree: 2. Decide the tree size

Method 1	Method 2
split tree nodes only if the decrease in sum-of-squares due to the split exceeds some threshold (C).	grow a large tree (T_0) (stop the splitting process only when some minimum node size (say 5) is reached) => tree then is pruned using cost-complexity criterion:
	$C_\alpha(T) = \sum_{m=1}^{ T } N_m Q_m(T) + \alpha T , T : \# \{terminal\ nodes\}$
Seemingly worthless split might lead to a very good split below it	If we have a lot of features, it may take time to grow T_0 .



Algorithm 8.1 *Building a Regression Tree*

Build trees $T_{\alpha_s}, s = 1, \dots, S$ for
each Candidate $\{\alpha_1, \alpha_2, \dots, \alpha_S\}$ $\longrightarrow$

1. Use recursive binary splitting to grow a large tree on the training data, stopping only when each terminal node has fewer than some minimum number of observations.
2. Apply cost complexity pruning to the large tree in order to obtain a sequence of best subtrees, as a function of α .

Pick $\hat{\alpha} \in \{\alpha_1, \alpha_2, \dots, \alpha_S\}$ $\longrightarrow$

3. Use K-fold cross-validation to choose α . That is, divide the training observations into K folds. For each $k = 1, \dots, K$:
 - (a) Repeat Steps 1 and 2 on all but the k th fold of the training data.
 - (b) Evaluate the mean squared prediction error on the data in the left-out k th fold, as a function of α .Average the results for each value of α , and pick α to minimize the average error.

$\hat{T} = T_{\hat{\alpha}}$ $\longrightarrow$

4. Return the subtree from Step 2 that corresponds to the chosen value of α .
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Reference:

The Elements of Statistical Learning, 2nd, Hastie & Tibshirani & Friedman